Finite public-source parity for a weighted genus-2 law

Source-bound review packet

25 October 2026

Protocol. The pinned author Magma projective functions `ADDDIVISOR` and their 18 dependencies were translated by a strict syntax-limited parser and evaluated in $\mathbf{F}_7$ and $\mathbf{F}_{11}$. All valid reduced divisors of $y^2 = x^5 + x + 2$ were enumerated independently of the compiled Rust enumerator. Every ordered pair was checked with unit weights and with $(z_1, w_1, z_2, w_2) = (2, 3, 3, 5)$. The author result was normalized and compared with compiled Rust; Rust was compared with independent Cantor arithmetic. A 1,024-row bounded gate preceded the complete pass.

Field, weight	Pairs	Author/Rust matches	Invalid weights	Undefined divisions
F7, 0	4,356	4,356	0	0
F7, 1	4,356	4,356	0	0
F11, 0	12,996	12,996	0	0
F11, 1	12,996	12,996	0	0

Observed. There are 17,352 ordered pairs and 34,704 weighted checks. All 34,704 author outputs matched Rust, and all 34,704 Rust outputs matched Cantor. There were zero invalid author output weights, zero undefined field divisions, and zero other invalid intermediates. The branch matrix has 108 joint path/label cells, covering 28 observed author call paths and 16 Rust labels. These are finite counts, not timing.
Evidence. `ROWS.csv` carries the four aggregate rows; `branch-matrix.csv` gives the joint paths and labels; `F7-weight*.csv.xz` and `F11-weight*.csv.xz` contain every standardized pair row. `SUMMARY.json` contains full raw digests, source hashes, and counts. `run.py` regenerates the data and checks it byte for byte. The author source SHA-256 is `7416a7fffaafff371738e7498a6795057372e1c477e0079297eebf2347c2e72a`; the Rust source SHA-256 is `5976e7f30b3e9c5c49a5fc0f4095404b7322832e4b02f467bc3cb7e0d9dcc8c0`.
Limit. Magma was unavailable, so the pinned Magma source was translated and evaluated, not run by Magma. The parser checks its exact source bytes and supported syntax, but this is not an independent Magma execution. One curve over each of two small fields cannot establish universal formula correctness, full Rust semantics, or the Picard-group bridge. The frozen 1,260-target costs are unchanged.